BIDYA BHUSHAN NANDA

BSP & Embedded Systems Engineer | ARM64 Linux, Bootloaders, UAV Systems

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SUMMARY

Electronics undergraduate (MAIT, 2029) who debugs where hardware meets software. Brought up Linux on a locked-down Qualcomm SDM429 tablet, root-caused secure-boot failures on real hardware, and built Kestrel, a Rust ground station for PX4 and ArduPilot. Also a full-stack engineer at AB Connectz and founder of Broccoli (about 100 users and a 5-star rating from 20+ reviews on Google Play, growing quickly).

TECHNICAL SKILLS

Languages: C/C++, Rust, Python, TypeScript, SQL
BSP & Linux: ARM64 kernel bring-up, bootloaders (S-Boot, LK2ND), device tree, initramfs, vbmeta / dm-verity, Qualcomm SDM429, Arch Linux ARM, LineageOS
Boot & Debug: fastboot, Odin, partition backup and checksum workflow, RAM-boot testing, kernel log analysis
UAV & Embedded: MAVLink v2, PX4, ArduPilot, SITL, Arduino, ESP32, Raspberry Pi, camera and sensor interfacing
Software & Tools: React Native, Node.js, NestJS, PostgreSQL, SQLite, Tauri, Git, Docker

PROJECTS

Zucchini Linux — Bare-Metal ARM64 Bring-Up (Samsung SM-T295) Oct 2026 – Present
C, Linux kernel, LK2ND, S-Boot, device tree, Arch Linux ARM, LineageOS | bidyabhushan.in/projects/zucchini-linux

- Dual-booted native Arch Linux ARM and LineageOS on a locked-down Qualcomm SDM429 tablet by reverse-engineering the S-Boot secure-boot chain and adding the LK2ND bootloader.
- Fixed a warm-reboot “SECURE CHECK FAIL: vbmeta” bootloop with a verification-disabled vbmeta image, after SHA-256-verified off-device backups of efs, persist and modem.
- Restored a clean boot to desktop by rebuilding the kernel with quota and dm-verity support after the Arch rootfs failed to mount in initramfs.
- Validated the LineageOS kernel and storage device tree (eMMC probed, display working) with RAM-boot guard images that keep eMMC read-only, so the Arch install was never at risk.
- Research paper: bidyabhushan.in/blog/sm-t295-linux-bringup

Kestrel — Custom Drone Ground Control Station Oct 2026 – Present
Rust, Tokio, Tauri v2, Svelte 5, MAVLink v2, PX4, ArduPilot, MediaMTX, PMTiles | bidyabhushan.in/projects/kestrel

- Isolates flight-critical MAVLink ingestion (CRC validation, heartbeat supervision) in a Rust/Tokio core, apart from the UI, and normalizes PX4 and ArduPilot flight modes.
- Blocks arming on stale or failing GPS, EKF, battery or RC-link data; commands stay pending until acknowledged (2.5 s timeout, never replayed after a reconnect).
- Built for 50 Hz telemetry with a lock-free state cache and binary IPC; video and offline maps run as isolated sidecars; tested on PX4 and ArduPilot SITL.

IoT QR-Code Scanner
Arduino, ESP32, Raspberry Pi 3B, C++, Python

- Built a QR-code scanner on Arduino and ESP32 (C++) and Raspberry Pi 3B (Python), covering camera capture, decoding and hardware I/O.

EXPERIENCE

AB Connectz — Remote Jun 2026 – Present
Full-Stack Engineer, Paid Apprenticeship (Intern, Jun – Jul 2026) Jul 2026 – Present

- Develop HOSHCARE, a health-records app in development: React Native / Expo, REST API, PostgreSQL with row-level security, and an AI pipeline that turns handwritten medical documents into structured records, tested on 500+ varied documents (lab reports, prescriptions, false data).
- Traced a silent auth failure to a breaking library update and a multipart upload failure to Expo’s fetch polyfill.

Broccoli — Founder & Developer (live on Google Play) Aug 2026 – Present

- Founded an offline-first Android nutrition tracker, newly in production with about 100 users, a 5-star rating from 20+ reviews, and quick growth. bidyabhushan.in/broccoli

EDUCATION

B.Tech, Electronics Engineering — MAIT, Delhi 2025 – 2029 (expected)
CGPA 8.5 / 10 | Aerostars UAV Society member (drones, IoT, embedded) | Systems skills self-taught and applied in hardware and product work.